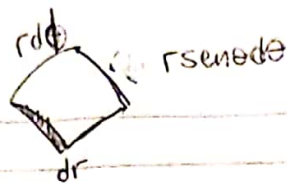
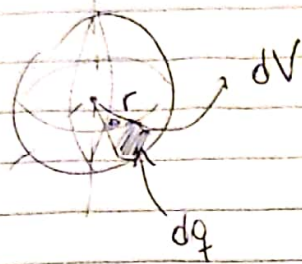


(6-)



$$dV = r^2 \sin \theta d\theta d\phi dr$$

$$\frac{dq}{dV} = \rho$$

$$\Rightarrow \rho dV = dq$$

$$Q = \int \rho dV$$

$$= \iiint \rho r^2 \sin \theta d\theta d\phi dr$$

$$= 8 \int_0^{2\pi} d\phi \int_0^{\pi} \sin \theta d\theta \int_0^R r^4 dr$$

$$= 8 (2\pi) (2) \frac{R^5}{5}$$

$$= \frac{32\pi}{5} (0.5)^5$$

$$= \cancel{0.2} \mu C.$$

$$= 0.628 \mu C.$$